Computer Networks LAB-01

Experiment :-

⇒ Study of LAN Topology, Network devices and Address configuration.

Objective :-

i> To study the LAN topology used in college campus.

ii> To identify networking device and understand their function.

iii> To analyze wired and wireless communication method used in the network.

iv) To measure speed of Network for wired and wireless connections.

v> To obtained MAC, IP Address and subnet mask.

Problem statement - 1

* Draw and identify LAN topology of college campus and clearly label all components such as rooter, switches and etc.

⇒ The college campus network observed this study consists of :

i> Layer-3 Core switches in the server Room.

ii> Managed by POE + switches.

iii> Wireless Access point (APs)

iv) Ethernet port in classroom and Laboratories

Network Topology :-

Hyborid Hierarchical Topology which combines:

i> Tree topology.

ii> Extended star topology.

iii> Ring topology (server room)

The network hierarchical layer

i> Core layer.

ii> Distribution layer

iii> Access layer.

i> Core Layer :-

Core layer contains server Room and server room have firewall, Router, 2-L3 switches, patch Rack, server, inverter etc.

ii> Distribution layer :-

- Distribution layer contains 4-switches.
- The 4 switches are :-

i> Main Build -> R7 (Room no 213, 2nd floor).

ii> main Build -> R1 (Room no-301, 3rd floor).

iii> main Build -> R6 (Room no-113, 1st floor).

iv> main Build -> R2 (Room no-201, 2nd floor)

- This 4-switches contains :-

i) 18 port gigabit Easy smart switch. with 16-Port. $\boxed{\text{TL-SG1218 MPE}}$

②

ii) $\boxed{SG3428XMP}$ -> 24 port Gigabit and 4-port 10GE SFP+, L2+ managed switches with 24 port POE+.

-> Now let's discuss these 2 above components.

i) TP-Link TL-SG1218MPE :-

- Likely used for: CC TV Cammera, wifi access point, etc.
- 18 port Gigabit "Easy smart" switch.
- 16 port support POE+

ii) TP-Link omada SG3428XMP :-

- Higher end managed Layer 2+ switch.
- 24 POE+ port, uplinks, 10G SFP.
- Usually acts as a main distribution for each floor.

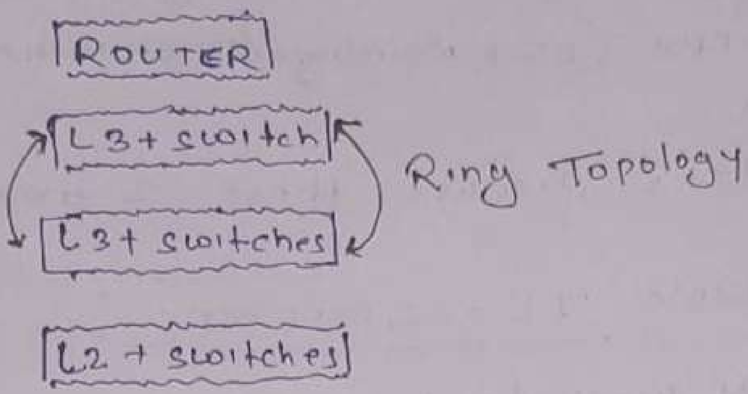
3) Access Layer :-

- Access layer contain 3-switches i.e
i) main Build R1 - [Room no ~~04~~ - 301, 3rd floor].
ii) main Build R4 - [Room no - 05, Ground floor]
iii) main Build R5 - [Room no - 18, Ground floor]

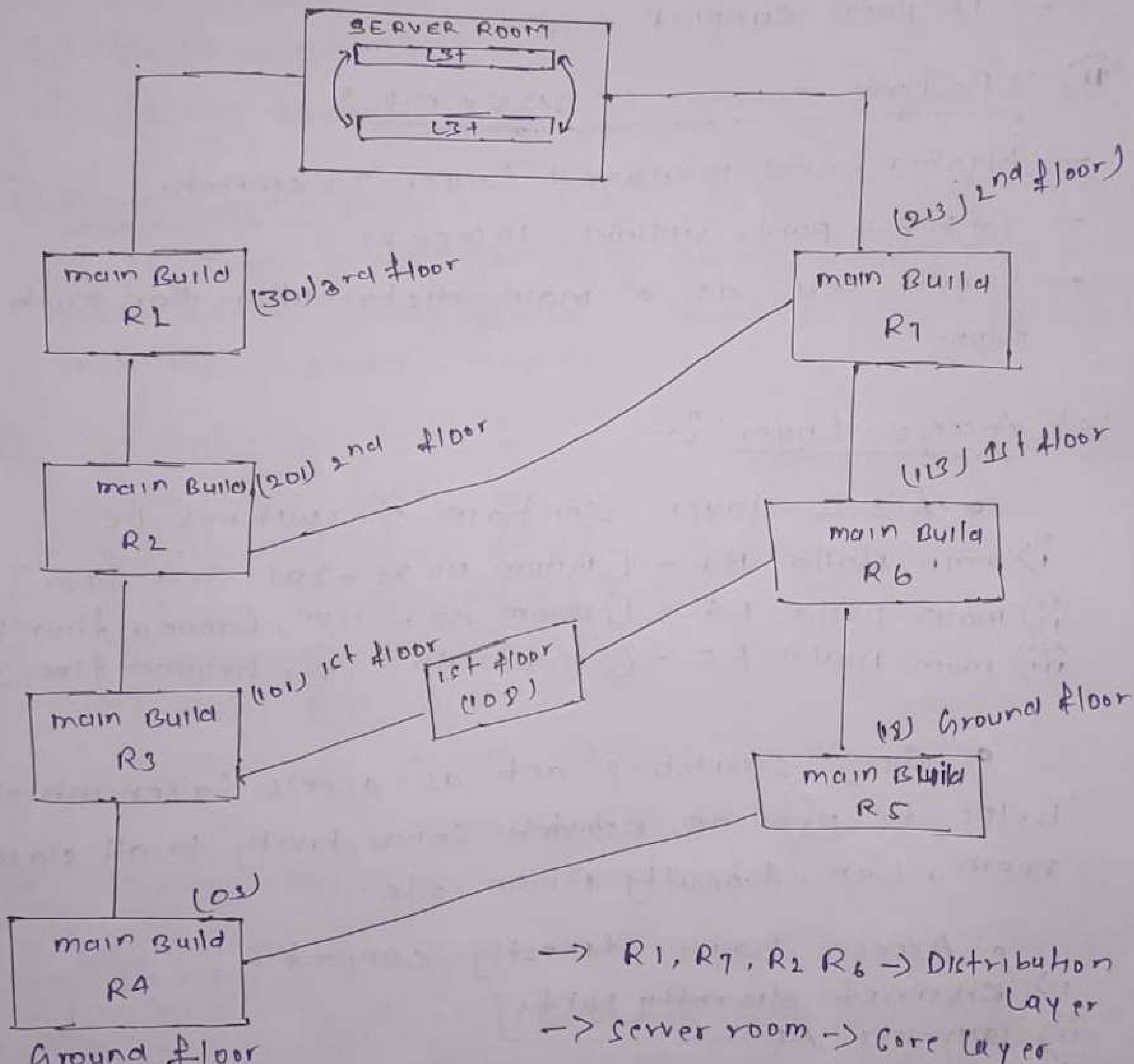
- Every switches act as access Layer which helps to provide network connectivity to all class room, Lab, faculty Room, etc.

- Access Layer directly connect.
i) Ethernet ~~[illegible]~~ port.
ii) wireless AP.
iii) Lab system.
iv) outdoor wireles AP (ex-018).

- SERVER ROOM Architecture / Topology.



TOPOLOGY

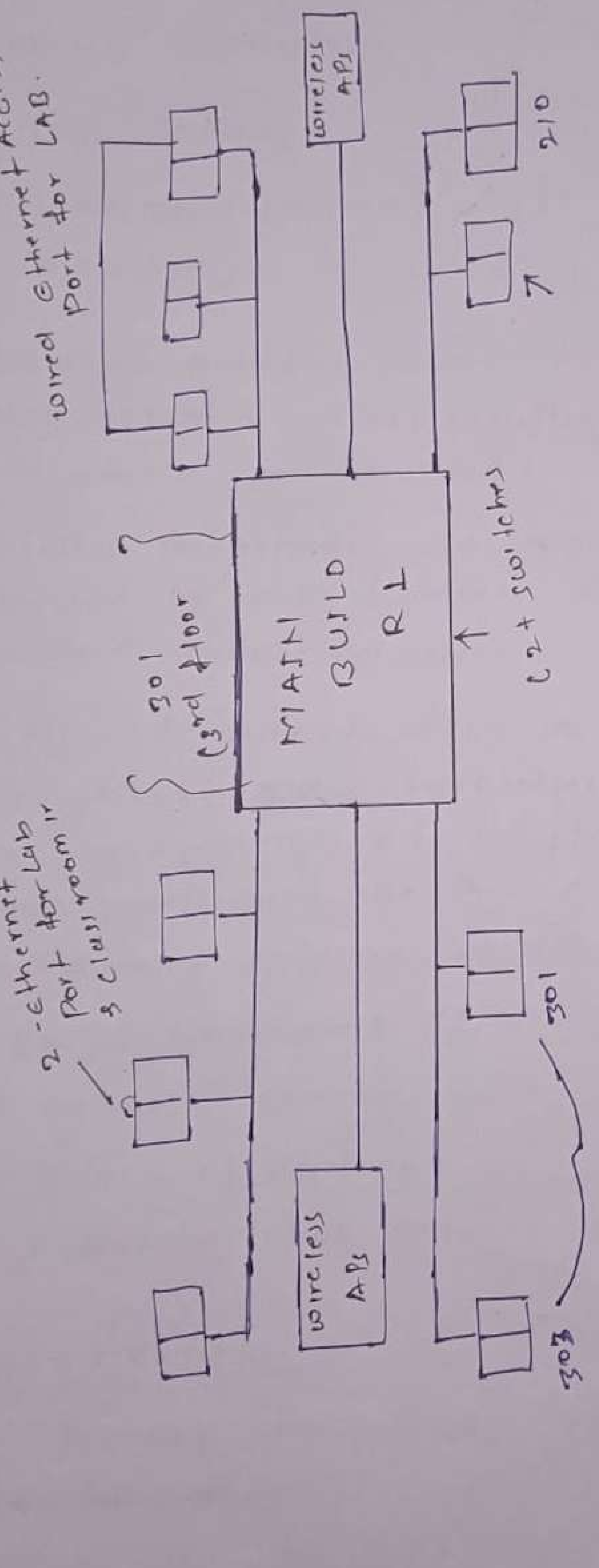


→ R1, R7, R2 R6 → Distribution Layer

→ Server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access Layer.

ACCESS LAYER SWITCH (3rd floor 301) :



Activity Monitor
All Processes

CPU | Memory | Energy | Disk | Network

Search

Process Name	Sent Byt...	Rcvd Bytes	Sent Packets	Rcvd Packets	PID	User
nsurlsessiond	6.7 MB	210.1 MB	92,380	1,71,148	413	_nsurlsessiond
mDNSResponder	258 KB	273 KB	627	718	496	_mdnsrespond
WhatsApp	119 KB	509 KB	663	906	2038	mr.beast
nsurlsessiond	73 KB	73 KB	201	180	625	mr.beast
apsd	22 KB	7 KB	61	51	365	root
mobileassetd	19 KB	100 KB	92	109	419	root
Google Chrome Helper	12 KB	17 KB	29	35	872	mr.beast
cloudd	10 KB	11 KB	21	25	609	mr.beast
parsecd	4 KB	18 KB	19	24	718	mr.beast
rtcreportingd	4 KB	6 KB	16	15	567	root
netbiosd	1 KB	2 KB	25	25	1967	_netbios
kernel_task	0 bytes	0 bytes	0	0	0	root
ScreenSharingSubscriber	0 bytes	0 bytes	0	0	1241	mr.beast
com.apple.accessibility.mediaaccessibilityd	0 bytes	0 bytes	0	0	1318	mr.beast
peakpowermanagerd	0 bytes	0 bytes	0	0	385	root
secinitd	0 bytes	0 bytes	0	0	539	_cmiodalassist
trustd	0 bytes	0 bytes	0	0	616	mr.beast
localizationswitcherd	0 bytes	0 bytes	0	0	693	mr.beast
com.apple.audio.SandboxHelper	0 bytes	0 bytes	0	0	1001	root
STExtractionService.privileged	0 bytes	0 bytes	0	0	1232	_nsurlsessiond
cfprefsd	0 bytes	0 bytes	0	0	1309	_softwareupda
corerepaird	0 bytes	0 bytes	0	0	376	root
ViewBridgeAuxiliary	0 bytes	0 bytes	0	0	530	root
containermanagerd	0 bytes	0 bytes	0	0	607	mr.beast
Dock Extra	0 bytes	0 bytes	0	0	684	mr.beast
CloudTelemetryService	0 bytes	0 bytes	0	0	761	mr.beast
ndoagent	0 bytes	0 bytes	0	0	992	mr.beast
studentd	0 bytes	0 bytes	0	0	1300	mr.beast
timed	0 bytes	0 bytes	0	0	367	_timed
mobileactivationd	0 bytes	0 bytes	0	0	521	root
UARPUpdaterServiceHID	0 bytes	0 bytes	0	0	598	_accessoryupc
intelligentroutingd	0 bytes	0 bytes	0	0	675	mr.beast
TextInputMenuAgent	0 bytes	0 bytes	0	0	752	mr.beast

Packets in:	4,21,993
Packets out:	1,99,581
Packets in/sec:	153
Packets out/sec:	77

PACKETS

Data received:	532.5 MB
Data sent:	28.3 MB
Data received/sec:	202 KB
Data sent/sec:	7 KB

```
Last login: Wed May 27 11:48:03 on ttys000
mr.beast@mrbeasts-MacBook-Air ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
	options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
	inet 127.0.0.1 netmask 0xff000000
	inet6 ::1 prefixlen 128
	inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
	nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
	options=400<CHANNEL_IO>
	ether 62:75:2f:09:e1:cf
	media: none
	status: inactive
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
	options=400<CHANNEL_IO>
	ether 62:75:2f:09:e1:ce
	media: none
	status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
	options=400<CHANNEL_IO>
	ether 62:75:2f:09:e1:ae
	nd6 options=201<PERFORMNUD,DAD>
	media: none
	status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
	options=400<CHANNEL_IO>
	ether 62:75:2f:09:e1:af
	nd6 options=201<PERFORMNUD,DAD>
	media: none
	status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
	options=460<TSO4,TSO6,CHANNEL_IO>
	ether 36:8f:94:9f:f6:40
	media: autoselect <full-duplex>
	status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
	options=460<TSO4,TSO6,CHANNEL_IO>
	ether 36:8f:94:9f:f6:44
	media: autoselect <full-duplex>
	status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
	options=63<RXCSUM,TXCSUM,TSO4,TSO6>
	ether 36:8f:94:9f:f6:40
	Configuration:
		id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
		maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
		root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
		ipfilter disabled flags 0x0
	member: en1 flags=3<LEARNING,DISCOVER>
	        ifmaxaddr 0 port 8 priority 0 path cost 0
	member: en2 flags=3<LEARNING,DISCOVER>
	        ifmaxaddr 0 port 9 priority 0 path cost 0
	nd6 options=201<PERFORMNUD,DAD>
	media: <unknown type>
	status: inactive
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500
	inet6 fe80::dd03:1d73:d925:5423%utun0 prefixlen 64 scopeid 0xd
	nd6 options=201<PERFORMNUD,DAD>
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
	inet6 fe80::f0f1:73d6:1e34:7ee9%utun1 prefixlen 64 scopeid 0xe
	nd6 options=201<PERFORMNUD,DAD>
ap1: flags=8822<BROADCAST,SMART,SIMPLEX,MULTICAST> mtu 1500
	options=400<CHANNEL_IO>
	ether ae:2b:22:c5:4e:8f
	nd6 options=201<PERFORMNUD,DAD>
	media: autoselect (none)
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
	options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
	ether 46:51:48:e5:5e:0b
	inet6 fe80::1cfe:8057:bb69:4b53%en0 prefixlen 64 secured scopeid 0xb
	inet 10.66.155.45 netmask 0xffffff00 broadcast 10.66.155.255
	inet6 2409:40e4:50:9ae0:1465:2ef:fb0f:499d prefixlen 64 autoconf secured
	inet6 2409:40e4:50:9ae0:65cc:8621:2603:1373 prefixlen 64 autoconf temporary
```

```
        media: autoselect (none)
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 46:51:48:e5:5e:0b
        inet6 fe80::1cfe:8057:bb69:4b53%en0 prefixlen 64 secured scopeid 0xb
        inet 10.66.155.45 netmask 0xffffff00 broadcast 10.66.155.255
        inet6 2409:40e4:50:9ae0:1465:2ef:fb0f:499d prefixlen 64 autoconf secured
        inet6 2409:40e4:50:9ae0:65cc:8621:2603:1373 prefixlen 64 autoconf temporary
        nat64 prefix 64:ff9b:: prefixlen 96
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
awdl0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 42:1d:bb:c0:af:93
        inet6 fe80::401d:bbff:fec0:af93%awdl0 prefixlen 64 scopeid 0xf
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
llw0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 42:1d:bb:c0:af:93
        inet6 fe80::401d:bbff:fec0:af93%llw0 prefixlen 64 scopeid 0x10
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (none)
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
        inet6 fe80::406d:ced3:e696:cc2d%utun2 prefixlen 64 scopeid 0x11
        nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
        inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
        nd6 options=201<PERFORMNUD,DAD>
mr.beast@mrbeasts-MacBook-Air ~ % mack
zsh: command not found: mack
mr.beast@mrbeasts-MacBook-Air ~ % ifconfig en1
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:8f:94:9f:f6:40
        media: autoselect <full-duplex>
        status: inactive
mr.beast@mrbeasts-MacBook-Air ~ %
```

Q. Difference b/w wired and wireless speed?

i) faster communication	i) slower communication.
ii) stable connection	ii) Affected by distance/ Interference.
iii) low latency	iii) Higher latency.
iv) Dedicated physical medium	iv) shared radio medium.

- Ethernet provides direct physical commonication using network cables, resulting in higher speed and stable performance.
- Wireless connections uses radio waves which are abbected by interference, distance, obstacles and multiple connected users.

Q. Role of subnet mask in network?

⇒ A subnet mask identifies the network portion and host portion of an IP address. It helps divices determine whether another device belongs to the same network or different network.

- It is important for!

i) network segment.

ii) Efficient routing.

iii) communication between devices.

OBSERVATION :-

During the study of the college network infrastructure:-

i) A centralized server room was used as network backbone.
ii) Two L3 switches were connected in redundant cyclic structure.
iii) Managed TP-Link PoE+ switches were used in network cabinets.

Q. Advantage of existing Network?
i) Scalable architecture.
ii) Centralized management.
iii) Supports PoE+ device.
iv) Easy expansion of floors and classroom.
v) Support both wireless and wired.
vi) Redundant link for failover.

CONCLUSION:-

The observed campus LAN follow a hybrid hierarchical topology consist of tree, Extended star, Ring. The network uses managed PoE switches, wireless access point, Ethernet port and centralized server infrastructure to provide efficient wired and wireless communication across the campus.